

Nộp bài tập homework ss10

Tên	Hoàng Duy Hướng
Lớp	PD16

1. Why don't we see hold violations in preCTS (placement)?

- Vì ở giai đoạn Placement/preCTS clock thật không tồn tại, tool chỉ giả định clock lý tưởng (ideal clock), vì vậy launch và capture clock chưa có các latency và skew. Nhưng sau CTS, clock được build là clock thật, tool sẽ bắt đầu tính toán latency và skew nên có xuất hiện các hold violate mới. Tùy vào positive skew hay negative skew nó sẽ tốt cho hold hay tệ cho hold

2. HW 1

2.1. How many violations found?

```
innovus 7> checkPlace
**Info: (IMPSP-307): Design contains fractional 1 cell.
Begin checking placement ... (start mem=1333.6M, init mem=1354.3M)
Overlapping with other instance:      10
*info: Placed = 50251                  (Fixed = 9486)
*info: Unplaced = 0
Placement Density:45.56%(630176/1383112)
Placement Density (including fixed std cells):46.22%(647088/1400024)
Finished checkPlace (total: cpu=0:00:01.9, real=0:00:02.0; vio checks: cpu=0:00:01.3, real=0:00:01.0; mem=1368.3M)
1
innovus 8> █
```

- Có 10 violations overlap ở design này

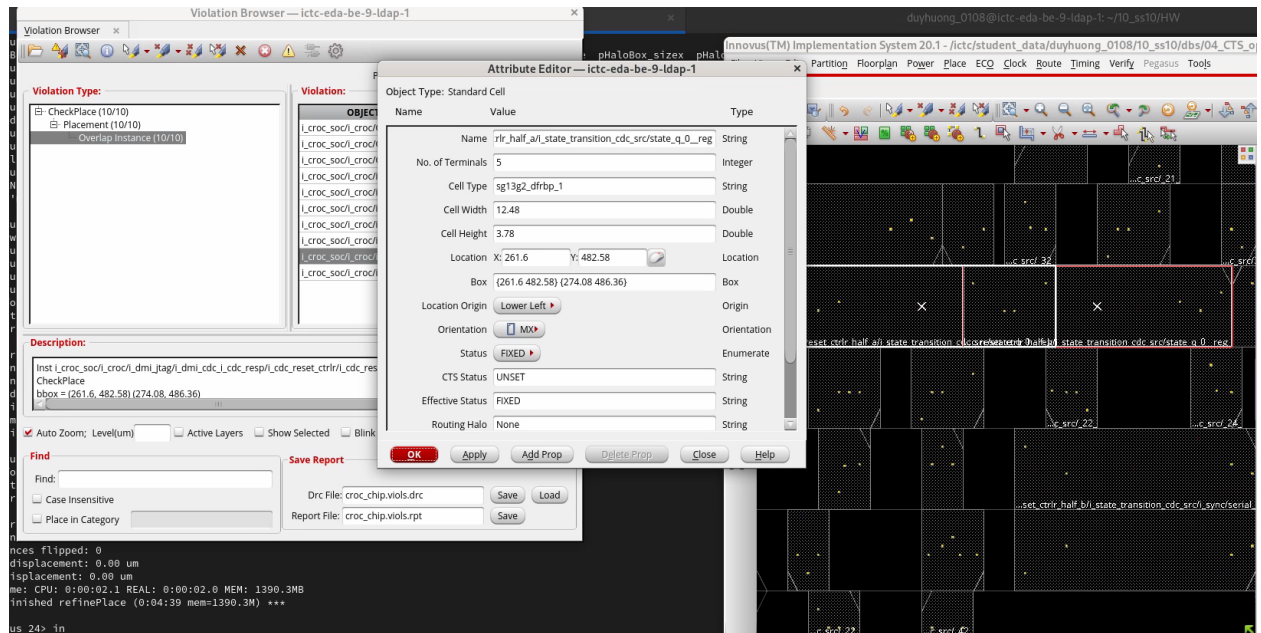
2.2. Try to fix it by “refinePlace -preserveRouting true”

```
innovus 8> refinePlace -preserveRouting true
**Info: (IMPSP-307): Design contains fractional 1 cell.
*** Starting refinePlace (0:02:39 mem=1372.4M) ***
Move report: Detail placement moves 0 insts, mean move: 0.00 um, max move: 0.00 um
Runtime: CPU: 0:00:02.0 REAL: 0:00:02.0 MEM: 1378.0MB
Summary Report:
Instances move: 0 (out of 40765 movable)
Instances flipped: 0
Mean displacement: 0.00 um
Max displacement: 0.00 um
Runtime: CPU: 0:00:02.1 REAL: 0:00:02.0 MEM: 1378.0MB
*** Finished refinePlace (0:02:41 mem=1378.0M) ***
1
```

- Không thể fix!

2.3. How many remain violations after fixing? Why tool cannot able to fix these violations?

- Vẫn còn 10 violations



- Tool không thể move các cell đó để fix overlap là vì status hiện tại của cell đang là fixed, nên tool không thể move các cell này!
- Quan trọng vẫn là thông số Effective status của cell, phải là Placed thì tool mới move được.

2.4. Try to fix and clean violations.

- Em đã chuyển các cell có thể chuyển trạng thái sang placed, và sau đó chạy lại “refinePlace” và tools đã tự động đặt lại cell để fix overlap

```

innovus 42> refinePlace
**Info: (IMPSP-307): Design contains fractional 1 cell.
*** Starting refinePlace (0:09:49 mem=1434.4M) ***
Move report: Detail placement moves 9 insts, mean move: 2.91 um, max move: 5.70 um
    Max move on inst (i_croc_soc/i_croc/i_dmi_jtag/i_dmi_cdc_i_cdc_resp/i_cdc_reset_ctrlr/i_cdc_reset_ctrlr_half_a/i_state_transition_cdc_src/38_): (267.84, 475.02) --> (265.92, 471.24)
    Runtime: CPU: 0:00:02.2 REAL: 0:00:03.0 MEM: 1434.4MB
Summary Report:
Instances move: 9 (out of 40770 movable)
Instances flipped: 0
Mean displacement: 2.91 um
Max displacement: 5.70 um (Instance: i_croc_soc/i_croc/i_dmi_jtag/i_dmi_cdc_i_cdc_resp/i_cdc_reset_ctrlr/i_cdc_reset_ctrlr_half_a/i_state_transition_cdc_src/38_) (267.84, 475.02) --> (265.92, 471.24)
    Length: 4 sites, height: 1 rows, site name: CoreSite, cell type: sg13g2_nor2_1
    Runtime: CPU: 0:00:02.4 REAL: 0:00:03.0 MEM: 1434.4MB
*** Finished refinePlace (0:09:51 mem=1434.4M) ***
1

```

- Kết quả sau khi khi chạy lại lệnh “checkPlace”

```

innovus 44> checkPlace
**Info: (IMPSP-307): Design contains fractional 1 cell.
Begin checking placement ... (start mem=1434.4M, init mem=1434.4M)
*info: Placed = 50251          (Fixed = 9481)
*info: Unplaced = 0
Placement Density:45.57%(630221/1383117)
Placement Density (including fixed std cells):46.22%(647088/1399984)
Finished checkPlace (total: cpu=0:00:02.2, real=0:00:02.0; vio checks: cpu=0:00:01.4, real=0:00:01.0; mem=1434.4M)
0
innovus 45> win
innovus 46>

```

- Report timing setup

```

-----
timeDesign Summary
-----
Setup views included:
func_view_wc

-----
| Setup mode | all | reg2reg | in2reg | reg2out | in2out | default | mem2reg | reg2mem |
-----
| WNS (ns): | -2.315 | -0.107 | 0.973 | -2.315 | N/A | 6.796 | 0.028 | 0.314 |
| TNS (ns): | -59.388 | -29.171 | 0.000 | -30.217 | N/A | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 829 | 794 | 0 | 35 | N/A | 0 | 0 | 0 |
| All Paths: | 10625 | 10058 | 261 | 35 | N/A | 2 | 1265 | 278 |
-----

-----
| DRVs | | Real | | Total |
| | | Nr nets(terms) | Worst Vio | Nr nets(terms) |
-----
| max_cap | 128 (128) | -0.027 | 167 (167) |
| max_tran | 0 (0) | 0.000 | 39 (78) |
| max_fanout | 1 (1) | -1 | 174 (174) |
| max_length | 0 (0) | 0 | 0 (0) |
-----

Density: 45.565%
Routing Overflow: 0.16% H and 0.53% V

-----
Reported timing to dir ./timingReports
Total CPU time: 2.11 sec
Total Real time: 2.0 sec
Total Memory Usage: 1426.40625 Mbytes
0
innovus 47> innovus 47>

```

- Không có thay đổi gì so với trước khi fixoverlap

3. HW 2

3.1. Check setup timing overall by: “timeDesign -postCTS -slackReport”

- Take snapshot of summary timing and put into report

```

-----
timeDesign Summary
-----
Setup views included:
func_view_wc

-----
| Setup mode | all | reg2reg | in2reg | reg2out | in2out | default | mem2reg | reg2mem |
-----
| WNS (ns): | -2.315 | -0.107 | 0.973 | -2.315 | N/A | 6.796 | 0.028 | 0.314 |
| TNS (ns): | -59.388 | -29.171 | 0.000 | -30.217 | N/A | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 829 | 794 | 0 | 35 | N/A | 0 | 0 | 0 |
| All Paths: | 10625 | 10058 | 261 | 35 | N/A | 2 | 1265 | 278 |
-----

-----
| DRVs | | Real | | Total |
| | | Nr nets(terms) | Worst Vio | Nr nets(terms) |
-----
| max_cap | 128 (128) | -0.027 | 167 (167) |
| max_tran | 0 (0) | 0.000 | 39 (78) |
| max_fanout | 1 (1) | -1 | 174 (174) |
| max_length | 0 (0) | 0 | 0 (0) |
-----

Density: 45.565%
Routing Overflow: 0.16% H and 0.53% V

-----
Reported timing to dir ./timingReports
Total CPU time: 2.77 sec
Total Real time: 2.0 sec
Total Memory Usage: 1426.414062 Mbytes
0
innovus 57> innovus 57>

```

- Ở path reg2reg thì có 794 path violations, WNS tệ nhất là -0.107 ns
- Ulti hiện tại là 45.565% (ulti hiện tại ổn)

3.2. Run: “report_timing -path_group reg2reg” and fix setup violate

- Use report_timing -from beginpoint -to endpoint -through <sizing cell> to confirm after fixing

```

8 Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_timer/counter_hi_1_target_reached_o_reg/CLK
9 Endpoint: i_croc_soc/i_croc/i_timer/counter_hi_1_target_reached_o_reg/D (*) checked with leading edge of 'clk_sys'
10 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15__reg/Q (*) triggered by leading edge of 'clk_sys'
11 Path Groups: (reg2reg)
12 Analysis View: func_view_wc
13 Other End Arrival Time 1.494
14 - Setup 0.155
15 + Phase Shift 10.000
16 + CPRR Adjustment 0.017
17 - Uncertainty 0.100
18 = Required Time 11.166
19 - Arrival Time 11.273
20 = Slack Time -0.107
21 Clock Rise Edge 0.000
22 + Clock Network Latency (Prop) 1.187
23 = Beginpoint Arrival Time 1.187
24 -----
25 Pin Cell Fanout Load Slew Delay Arrival Pin
26 Location
27 -----
28 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15__reg/CLK - 28 0.241 0.248 - 1.187 (1190.88, 635.88)
29 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15__reg/Q sg13g2_dfrbp_1 1 0.011 0.091 0.454 1.641 (1200.48, 636.72)
30 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OCP0154_i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_4 5 0.048 0.080 0.102 1.743 (1202.46, 636.72)
31 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP0155_i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_2 3 0.036 0.125 0.134 1.877 (1224.08, 636.72)
32 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP0133_i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_8 8 0.121 0.107 0.122 1.999 (1230.72, 606.48)
33 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP0816_n/Y sg13g2_inv_16 7 0.112 0.075 0.095 2.094 (1243.68, 601.86)
34 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/4469_7/X sg13g2_mux4_1 1 0.004 0.069 0.383 2.477 (1340.16, 585.48)
35 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_RC_1419_0/Y sg13g2_a220i_1 1 0.006 0.099 0.137 2.614 (1340.64, 582.12)
36 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_341_0/Y sg13g2_nor2b_2 1 0.018 0.151 0.246 2.860 (1331.04, 577.92)
37 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8673_0/Y sg13g2_nand3_1 1 0.009 0.170 0.202 3.062 (1252.80, 587.16)
38 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OFC5501_i_ibex_ex_block_1_multdiv_operand_a_1_11/X sg13g2_buf_8 1 0.066 0.070 0.203 3.265 (1256.64, 585.48)
39 i_croc_soc/i_croc/i_core_wrap/FE_RC_724_0/Y sg13g2_nand2b_2 1 0.029 0.119 0.125 3.390 (1173.68, 859.32)
40 i_croc_soc/i_croc/i_core_wrap/FE_RC_4072_0/Y sg13g2_nand2b_2 1 0.015 0.108 0.140 3.530 (1075.20, 884.52)
41 i_croc_soc/i_croc/i_core_wrap/FE_RC_4076_0/Y sg13g2_a210i_2 1 0.012 0.120 0.154 3.684 (1019.52, 885.78)
42 i_croc_soc/i_croc/i_core_wrap/FE_OCP05254_1065/Y sg13g2_inv_4 3 0.039 0.076 0.103 3.786 (1022.88, 886.20)
43 i_croc_soc/i_croc/i_core_wrap/FE_RC_8768_0/Y sg13g2_a210i_2 2 0.013 0.121 0.149 3.936 (936.48, 832.86)
44 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_199_0_dup/Y sg13g2_inv_2 1 0.007 0.045 0.072 4.008 (935.52, 827.40)
45 i_croc_soc/i_croc/i_core_wrap/FE_RC_8441_0/Y sg13g2_nor3_2 2 0.009 0.146 0.139 4.147 (935.52, 819.84)
46 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1491_0/Y sg13g2_a210i_2 2 0.019 0.125 0.121 4.268 (934.08, 817.74)
47 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1676_0/Y sg13g2_nor3_2 1 0.005 0.114 0.169 4.437 (940.32, 804.72)
48 i_croc_soc/i_croc/i_core_wrap/FE_RC_3627_0/Y sg13g2_nor2b_2 1 0.012 0.108 0.222 4.658 (936.48, 797.16)
49 i_croc_soc/i_croc/i_core_wrap/FE_OCP0106_FE_RN_1487_0/Y sg13g2_inv_4 6 0.039 0.073 0.099 4.757 (939.36, 797.16)
50 i_croc_soc/i_croc/i_core_wrap/FE_RC_185_0/Y sg13g2_a210i_2 4 0.018 0.164 0.178 4.927 (804.24, 795.06)
51 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1000_0/Y sg13g2_o21a1_1 1 0.006 0.112 0.155 5.082 (890.40, 774.48)
52 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1275_0/Y sg13g2_nand2_2 2 0.008 0.055 0.079 5.160 (886.08, 771.12)
53 i_croc_soc/i_croc/i_core_wrap/FE_RC_8412_0/Y sg13g2_nand3_1 1 0.009 0.166 0.177 5.337 (885.60, 768.60)
54 i_croc_soc/i_croc/i_core_wrap/FE_RC_3620_0/Y sg13g2_a210i_1 2 0.010 0.108 0.163 5.500 (903.84, 767.34)
55 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1748_0/Y sg13g2_a210i_2 2 0.031 0.182 0.205 5.705 (907.68, 767.34)

```

- Sau khi sử dụng các phương pháp để fix setup violate timing thì path đã MET

```

8 Path 1: MET Setup Check with Pin i_croc_soc/i_croc/i_timer/counter_hi_1_target_reached_o_reg/CLK
9 Endpoint: i_croc_soc/i_croc/i_timer/counter_hi_1_target_reached_o_reg/D (*) checked with leading edge of 'clk_sys'
10 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15__reg/Q (*) triggered by leading edge of 'clk_sys'
11 Path Groups: (reg2reg)
12 Analysis View: func_view_wc
13 Other End Arrival Time 1.494
14 - Setup 0.155
15 + Phase Shift 10.000
16 + CPRR Adjustment 0.017
17 - Uncertainty 0.100
18 = Required Time 11.166
19 - Arrival Time 11.160
20 = Slack Time 0.006
21 Clock Rise Edge 0.000
22 + Clock Network Latency (Prop) 1.187
23 = Beginpoint Arrival Time 1.187
24 -----
25 Pin Cell Fanout Load Slew Delay Arrival Pin
26 Location
27 -----
28 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15__reg/CLK - 28 0.241 0.248 - 1.187 (1190.88, 635.88)
29 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15__reg/Q -> sg13g2_dfrbp_1 1 0.011 0.091 0.454 1.641 (1200.48, 636.72)
30 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OCP0154_i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_4 5 0.064 0.098 0.121 1.761 (1202.40, 636.72)
31 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP0155_i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_8 3 0.058 0.070 0.085 1.846 (1224.00, 636.72)
32 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP0133_i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_16 8 0.121 0.062 0.072 1.918 (1230.72, 606.22)
33 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP0816_n/Y -> sg13g2_inv_16 7 0.089 0.057 0.076 1.994 (1243.68, 601.86)
34 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP0822_n/Y sg13g2_inv_8 4 0.066 0.060 0.084 2.078 (1352.16, 591.36)
35 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP0824_n/Y sg13g2_inv_8 3 0.025 0.038 0.059 2.136 (1351.20, 583.80)
36 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/4469_7/X sg13g2_mux4_1 1 0.004 0.064 0.336 2.472 (1359.84, 585.48)
37 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_5330_0/Y sg13g2_a220i_1 1 0.003 0.086 0.111 2.583 (1365.60, 582.12)
38 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_776_0/Y sg13g2_nor2b_2 1 0.022 0.174 0.255 2.838 (1362.72, 583.80)
39 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_2753_0/Y sg13g2_nand3_1 1 0.005 0.126 0.166 3.004 (1259.52, 589.68)
40 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OCP0206_i_ibex_ex_block_1_multdiv_operand_a_1_1/X sg13g2_buf_4 1 0.066 0.108 0.215 3.219 (1261.92, 591.36)
41 i_croc_soc/i_croc/i_core_wrap/FE_RC_752_0/Y sg13g2_nand2b_2 1 0.008 0.056 0.090 3.309 (1172.16, 866.88)
42 i_croc_soc/i_croc/i_core_wrap/FE_RC_751_0/Y sg13g2_nand2_2 1 0.007 0.062 0.078 3.387 (1159.20, 869.40)
43 i_croc_soc/i_croc/i_core_wrap/FE_RC_389_0/Y sg13g2_a210i_2 1 0.012 0.116 0.132 3.519 (1164.96, 865.62)
44 i_croc_soc/i_croc/i_core_wrap/FE_OCP0370_FE_RN_206_0/Y sg13g2_inv_4 8 0.031 0.118 0.142 3.662 (1168.32, 865.20)
45 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1545_0/Y sg13g2_xnor2_1 1 0.005 0.125 0.175 3.837 (1010.40, 770.70)
46 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1567_0/Y sg13g2_nand3_1 1 0.007 0.154 0.199 4.035 (1008.00, 778.68)
47 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1542_0/Y sg13g2_nor2_2 1 0.006 0.078 0.134 4.169 (1003.20, 782.04)
48 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1543_0/Y sg13g2_inv_4 6 0.019 0.056 0.078 4.247 (1006.08, 782.04)
49 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1346_0/Y sg13g2_nand3_1 1 0.007 0.078 0.083 4.331 (968.64, 793.80)
50 i_croc_soc/i_croc/i_core_wrap/FE_RC_3628_0/Y sg13g2_nand2_2 1 0.011 0.077 0.113 4.444 (966.24, 798.84)
51 i_croc_soc/i_croc/i_core_wrap/FE_RC_3627_0/Y sg13g2_nor2b_2 1 0.012 0.108 0.119 4.562 (936.48, 797.16)
52 i_croc_soc/i_croc/i_core_wrap/FE_OCP0106_FE_RN_1487_0/Y sg13g2_inv_4 6 0.039 0.072 0.099 4.661 (939.36, 797.16)
53 i_croc_soc/i_croc/i_core_wrap/FE_RC_185_0/Y sg13g2_a210i_2 4 0.018 0.164 0.170 4.831 (804.24, 795.06)
54 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1000_0/Y sg13g2_o21a1_1 1 0.006 0.112 0.155 4.986 (890.40, 774.48)

```

3.3. Fix hold violations

- Use option -early to report hold timing
- Có 3 path violate hold timing: timeDesign -postCTS -slackRepot -hold

```
-----
timeDesign Summary
-----

Hold views included:
func_view_bc

+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Hold mode | all | reg2reg | in2reg | reg2out | in2out | default | mem2reg | reg2mem |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| WNS (ns): | -1.076 | -1.076 | 0.374 | 2.498 | N/A | 0.000 | 2.157 | 0.066 |
| TNS (ns): | -1.244 | -1.244 | 0.000 | 0.000 | N/A | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 3 | 3 | 0 | 0 | N/A | 0 | 0 | 0 |
| All Paths: | 10405 | 10058 | 43 | 35 | N/A | 0 | 1265 | 278 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+

Density: 45.566%
Routing Overflow: 0.16% H and 0.53% V
-----

Reported timing to dir ./timingReports
Total CPU time: 14.97 sec
Total Real time: 14.0 sec
Total Memory Usage: 1619.773438 Mbytes
0
```

- Path 1 hold violate hold timing

```
Path 1: VIOLATED Hold Check with Pin i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_187__reg/CLK
Endpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_187__reg/D (v) checked with leading edge of 'clk_sys'
Beginpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_251__reg/Q_N (v) triggered by leading edge of 'clk_sys'
Path Groups: {reg2reg}
Analysis View: func_view_bc
Other End Arrival Time 1.531
+ Hold -0.019
+ Phase Shift 0.000
+ CPRP Adjustment 0.007
+ Uncertainty 0.100
= Required Time 1.605
Arrival Time 0.528
Slack Time -1.076
Clock Rise Edge 0.000
+ Drive Adjustment 0.122
= Beginpoint Arrival Time 0.122
Timing Path:

Pin Cell Fanout Load Slew Delay Arrival Pin
Time Location

clk_i - 1 0.196 0.167 - 0.122 (220.00, 1638.50)
pad_clk_i/p2c sg13g2_IOPadIn 1 0.027 0.048 0.052 0.174 (220.03, 1460.14)
i_croc_soc/CTS_cld_buf_00531/X sg13g2_buf_16 4 0.236 0.031 0.059 0.233 (241.92, 1406.16)
i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_251__reg/Q_N -> sg13g2_dfrbp_1 2 0.011 0.045 0.232 0.465 (456.00, 801.78)
i_croc_soc/i_croc/i_gpio/_1082_/X -> sg13g2_and3_1 1 0.003 0.014 0.063 0.528 (451.68, 797.58)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_187__reg/D -> sg13g2_dfrbp_1 - 0.003 0.014 0.000 0.528 (454.56, 797.58)

Clock Rise Edge 0.000
+ Drive Adjustment 0.129
= Beginpoint Arrival Time 0.129
Other End Path:

Pin Cell Fanout Load Slew Delay Arrival Pin
Time Location

clk_i - 1 0.240 0.171 - 0.129 (220.00, 1638.50)
pad_clk_i/p2c sg13g2_IOPadIn 1 0.027 0.048 0.052 0.181 (220.03, 1460.14)
i_croc_soc/CTS_cld_buf_00531/X sg13g2_buf_16 4 0.236 0.031 0.059 0.240 (241.92, 1406.16)
i_croc_soc/CTS_ccl_a_buf_00496/X sg13g2_buf_16 7 0.231 0.050 0.167 0.407 (516.48, 922.32)
i_croc_soc/CTS_ccl_a_buf_00486/X sg13g2_buf_16 32 0.289 0.052 0.083 0.489 (552.48, 922.32)
i_croc_soc/i_croc/CTS_ccl_a_buf_00183/X sg13g2_buf_2 32 0.155 0.222 0.198 0.688 (432.48, 918.12)
i_croc_soc/i_croc/i_gpio/ictc_postCTS_setupFE_ECOC09_CTS_32/X sg13g2_buf_2 1 0.004 0.023 0.135 0.823 (468.48, 795.48)
i_croc_soc/i_croc/i_gpio/ictc_postCTS_setupFE_ECOC1_CTS_32/X sg13g2_buf_2 1 0.004 0.014 0.046 0.869 (470.88, 803.04)
i_croc_soc/i_croc/i_gpio/ictc_postCTS_setupFE_ECOC2_CTS_32/X sg13g2_buf_2 1 0.005 0.015 0.045 0.914 (473.28, 803.04)
```

- Add buffer và down size cell ở launch path để thêm delay

```

Path 1: MET Hold Check with Pin i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_187__reg/CLK
Endpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_187__reg/D (*) checked with leading edge of 'clk_sys'
Beginpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_251__reg/Q_N (*) triggered by leading edge of 'clk_sys'
Path Groups: {reg2reg}
Analysis View: func_view_bc
Other End Arrival Time 1.601
+ Hold -0.040
+ Phase Shift 0.000
+ CPPR Adjustment 0.007
+ Uncertainty 0.100
= Required Time 1.654
Arrival Time 1.714
Slack Time 0.060
Clock Rise Edge 0.000
+ Drive Adjustment 0.122
= Beginpoint Arrival Time 0.122
Timing Path:

```

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.196	0.167	-	0.122	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	1	0.011	0.032	0.046	0.168	(220.03, 1460.14)
i_croc_soc/CTS_c1d_buf_00531/X	sg13g2_buf_2	4	0.226	0.335	0.188	0.356	(232.32, 1407.84)
i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_251__reg/Q_N ->	sg13g2_dfrbp_1	2	0.012	0.047	0.290	0.646	(439.20, 821.10)
i_croc_soc/i_croc/i_gpio/i_1082/X ->	sg13g2_and3_1	1	0.021	0.067	0.125	0.771	(437.28, 779.94)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF/X	sg13g2_buf_16	1	0.019	0.016	0.065	0.836	(489.12, 806.40)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_1/X	sg13g2_buf_16	1	0.021	0.013	0.044	0.880	(459.36, 816.48)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_2/X	sg13g2_buf_16	1	0.021	0.013	0.044	0.924	(460.32, 819.00)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_3/X	sg13g2_buf_16	1	0.021	0.013	0.044	0.968	(461.28, 821.52)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_4/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.012	(462.24, 824.04)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_5/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.056	(463.20, 826.56)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_6/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.100	(464.16, 829.08)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_7/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.144	(465.12, 831.60)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_8/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.188	(466.08, 834.12)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_9/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.232	(467.04, 836.64)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_10/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.276	(468.00, 839.16)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_11/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.320	(468.96, 841.68)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_12/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.364	(469.92, 844.20)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_13/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.408	(470.88, 846.72)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_14/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.452	(471.84, 849.24)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_15/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.496	(472.80, 851.76)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_16/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.540	(473.76, 854.28)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_17/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.584	(474.72, 856.80)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_18/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.628	(475.68, 859.32)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_19/X	sg13g2_buf_16	1	0.021	0.013	0.044	1.672	(476.64, 861.84)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_20/X	sg13g2_buf_16	1	0.002	0.010	0.041	1.714	(477.60, 864.36)

- Còn cách nữa là detur net mà tool lúc này anh giảng viên có đề cập bị lỗi highlight path nên tạm thời không dùng cách này được!

- Path 2
- Path này có âm slack cao hơn nhưng em đang fix giữa chừng quên chụp ạ

```

Path 1: VIOLATED Hold Check with Pin i_croc_soc/i_croc/i_gpio/i_reg_file_reg2hw_190__reg/CLK
Endpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_reg2hw_190__reg/D - (v) checked with leading edge of 'clk_sys'
Beginpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_251__reg/Q_N (v) triggered by leading edge of 'clk_sys'
Path Groups: {reg2reg}
Analysis View: func_view_bc
Other End Arrival Time 0.776
+ Hold -0.007
+ Phase Shift 0.000
+ CPPR Adjustment 0.007
+ Uncertainty 0.100
= Required Time 0.862
Arrival Time 0.736
Slack Time -0.126
Clock Rise Edge 0.000
+ Drive Adjustment 0.122
= Beginpoint Arrival Time 0.122
Timing Path:

```

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.196	0.167	-	0.122	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	1	0.011	0.032	0.046	0.168	(220.03, 1460.14)
i_croc_soc/CTS_c1d_buf_00531/X	sg13g2_buf_2	4	0.226	0.335	0.188	0.356	(232.32, 1407.84)
i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_251__reg/Q_N ->	sg13g2_dfrbp_1	2	0.011	0.045	0.294	0.650	(439.20, 821.10)
i_croc_soc/i_croc/i_gpio/i_1005/Y ->	sg13g2_nand3_1	1	0.007	0.036	0.043	0.693	(468.96, 801.36)
i_croc_soc/i_croc/i_gpio/i_1007/Y	sg13g2_o21a1_1	1	0.004	0.027	0.043	0.736	(465.12, 787.92)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg2hw_190__reg/D ->	sg13g2_dfrbp_2	-	0.004	0.027	0.000	0.736	(471.84, 782.46)

```

Clock Rise Edge 0.000
+ Drive Adjustment 0.129
= Beginpoint Arrival Time 0.129
Other End Path:

```

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.240	0.171	-	0.129	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	1	0.011	0.032	0.046	0.175	(220.03, 1460.14)
i_croc_soc/CTS_c1d_buf_00531/X	sg13g2_buf_2	4	0.226	0.335	0.188	0.363	(232.32, 1407.84)
i_croc_soc/CTS_ccl_buf_00469/X	sg13g2_buf_8	7	0.240	0.095	0.252	0.614	(509.28, 924.00)
i_croc_soc/i_croc/CTS_ccl_a_buf_00469/X	sg13g2_buf_16	30	0.294	0.051	0.103	0.718	(630.72, 922.32)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg2hw_190__reg/CLK	sg13g2_dfrbp_2	-	0.294	0.128	0.058	0.776	(465.60, 782.46)

- Path này em tăng size cell ở capture giảm delay và add buf ở launch để tăng delay

```

Path 1: MET Hold Check with Pin i_croc_soc/i_croc/i_gpio/i_reg_file_reg2hw_190__reg/CLK
Endpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_reg2hw_190__reg/D (v) checked with leading edge of 'clk_sys'
Beginpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_251__reg/Q_N (v) triggered by leading edge of 'clk_sys'
Path Groups: {reg2reg}
Analysis View: func_view_bc
Other End Arrival Time 0.776
+ Hold -0.003
+ Phase Shift 0.000
- CPDR Adjustment 0.007
+ Uncertainty 0.100
= Required Time 0.866
Arrival Time 0.906
Slack Time 0.040
Clock Rise Edge 0.000
+ Drive Adjustment 0.122
= Beginpoint Arrival Time 0.122
Timing Path:

```

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.196	0.167	-	0.122	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	1	0.011	0.032	0.046	0.168	(220.03, 1460.14)
i_croc_soc/CTS_cld_buf_00531/X	sg13g2_buf_2	4	0.226	0.335	0.188	0.356	(232.32, 1407.84)
i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_251__reg/Q_N ->	sg13g2_dfrbp_1	2	0.011	0.045	0.204	0.650	(439.20, 821.10)
i_croc_soc/i_croc/i_gpio/i_1605/Y ->	sg13g2_nand3_1	1	0.007	0.036	0.043	0.693	(468.96, 801.36)
i_croc_soc/i_croc/i_gpio/i_1607/Y	sg13g2_o2lai_1	1	0.020	0.088	0.092	0.785	(465.12, 787.92)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_0/X	sg13g2_buf_16	1	0.020	0.017	0.075	0.860	(466.08, 790.44)
i_croc_soc/i_croc/i_gpio/ECO_INST_BUF_01/X	sg13g2_buf_16	1	0.004	0.011	0.046	0.906	(467.04, 792.96)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg2hw_190__reg/D ->	sg13g2_dfrbp_2	-	0.004	0.011	0.000	0.906	(471.84, 782.46)

```

Clock Rise Edge 0.000
+ Drive Adjustment 0.129
= Beginpoint Arrival Time 0.129
Other End Path:

```

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.240	0.171	-	0.129	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	1	0.011	0.032	0.046	0.175	(220.03, 1460.14)
i_croc_soc/CTS_cld_buf_00531/X	sg13g2_buf_2	4	0.226	0.335	0.188	0.363	(232.32, 1407.84)
i_croc_soc/CTS_ccl_buf_00496/X	sg13g2_buf_8	7	0.240	0.095	0.252	0.614	(509.28, 924.00)
i_croc_soc/i_croc/CTS_ccl_a_buf_00469/X	sg13g2_buf_16	30	0.294	0.051	0.103	0.718	(630.72, 922.32)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg2hw_190__reg/CLK	sg13g2_dfrbp_2	-	0.294	0.128	0.058	0.776	(465.60, 782.46)

- Path 3

```

Path 1: VIOLATED Hold Check with Pin i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_168__reg/CLK
Endpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_168__reg/D (v) checked with leading edge of 'clk_sys'
Beginpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_232__reg/Q_N (v) triggered by leading edge of 'clk_sys'
Path Groups: {reg2reg}
Analysis View: func_view_bc
Other End Arrival Time 0.920
+ Hold 0.000
+ Phase Shift 0.000
- CPDR Adjustment 0.007
+ Uncertainty 0.100
= Required Time 1.031
Arrival Time 1.015
Slack Time -0.016
Clock Rise Edge 0.000
+ Drive Adjustment 0.122
= Beginpoint Arrival Time 0.122
Timing Path:

```

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.196	0.167	-	0.122	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	1	0.011	0.032	0.046	0.168	(220.03, 1460.14)
i_croc_soc/CTS_cld_buf_00531/X	sg13g2_buf_2	4	0.226	0.335	0.188	0.356	(232.32, 1407.84)
i_croc_soc/CTS_ccl_buf_00496/X	sg13g2_buf_8	7	0.240	0.095	0.252	0.608	(509.28, 924.00)
i_croc_soc/CTS_ccl_a_buf_00486/X	sg13g2_buf_16	32	0.305	0.057	0.101	0.708	(552.48, 922.32)
i_croc_soc/i_croc/CTS_ccl_a_buf_00183/X	sg13g2_buf_16	32	0.155	0.040	0.090	0.798	(432.48, 919.80)
i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_232__reg/Q_N ->	sg13g2_dfrbp_1	2	0.009	0.040	0.154	0.952	(422.88, 862.26)
i_croc_soc/i_croc/i_gpio/i_1093/X ->	sg13g2_and3_1	1	0.004	0.016	0.062	1.015	(412.32, 870.66)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_168__reg/D ->	sg13g2_dfrbp_1	-	0.004	0.016	0.000	1.015	(405.12, 873.18)

```

Clock Rise Edge 0.000
+ Drive Adjustment 0.129
= Beginpoint Arrival Time 0.129
Other End Path:

```

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.240	0.171	-	0.129	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	1	0.011	0.032	0.046	0.175	(220.03, 1460.14)
i_croc_soc/CTS_cld_buf_00531/X	sg13g2_buf_2	4	0.226	0.335	0.188	0.363	(232.32, 1407.84)
i_croc_soc/CTS_ccl_buf_00496/X	sg13g2_buf_8	7	0.240	0.095	0.252	0.614	(509.28, 924.00)
i_croc_soc/CTS_ccl_a_buf_00486/X	sg13g2_buf_16	32	0.305	0.057	0.101	0.715	(552.48, 922.32)
i_croc_soc/i_croc/CTS_ccl_a_buf_00331/X	sg13g2_buf_2	32	0.147	0.206	0.206	0.921	(421.44, 903.00)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_168__reg/CLK	sg13g2_dfrbp_1	-	0.147	0.207	0.008	0.929	(407.04, 873.60)

- Path này em chỉ cần up size cell ở capture path là path đã MET

```

Path 1: MET Hold Check with Pin i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_168__reg/CLK
Endpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_168__reg/D (v) checked with leading edge of 'clk_sys'
Beginpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_232__reg/Q_N (v) triggered by leading edge of 'clk_sys'
Path Groups: {reg2reg}
Analysis View: func_view_bc
Other End Arrival Time 0.642
+ Hold -0.015
+ Phase Shift 0.000
- CPFR Adjustment 0.007
+ Uncertainty 0.100
= Required Time 0.720
Arrival Time 0.840
Slack Time 0.120
Clock Rise Edge 0.000
+ Drive Adjustment 0.122
= Beginpoint Arrival Time 0.122
Timing Path:
-----
Pin Cell Fanout Load Slew Delay Arrival Pin
Time Location
-----
clk_i - 1 0.196 0.167 - 0.122 (220.00, 1638.50)
pad_clk_i/p2c sg13g2_IOPadIn 1 0.018 0.038 0.048 0.170 (220.03, 1460.14)
i_croc_soc/CTS_cld_buf_00531/X sg13g2_buf_8 4 0.226 0.063 0.062 0.232 (234.72, 1407.84)
i_croc_soc/CTS_ccl_buf_00496/X sg13g2_buf_8 7 0.240 0.090 0.197 0.429 (509.28, 924.00)
i_croc_soc/CTS_ccl_a_buf_00486/X sg13g2_buf_16 32 0.321 0.060 0.101 0.530 (552.48, 922.32)
i_croc_soc/i_croc/CTS_ccl_a_buf_00183/X sg13g2_buf_16 32 0.155 0.040 0.094 0.624 (432.48, 919.80)
i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_232__reg/Q_N -> sg13g2_dfrbp_1 2 0.009 0.040 0.154 0.778 (422.88, 862.26)
i_croc_soc/i_croc/i_gpio/_1093_/X -> sg13g2_and3_1 1 0.004 0.016 0.062 0.840 (412.32, 870.66)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_168__reg/D -> sg13g2_dfrbp_1 - 0.004 0.016 0.000 0.840 (405.12, 873.18)
-----
Clock Rise Edge 0.000
+ Drive Adjustment 0.129
= Beginpoint Arrival Time 0.129
Other End Path:
-----
Pin Cell Fanout Load Slew Delay Arrival Pin
Time Location
-----
clk_i - 1 0.240 0.171 - 0.129 (220.00, 1638.50)
pad_clk_i/p2c sg13g2_IOPadIn 1 0.018 0.039 0.048 0.177 (220.03, 1460.14)
i_croc_soc/CTS_cld_buf_00531/X sg13g2_buf_8 4 0.226 0.063 0.062 0.239 (234.72, 1407.84)
i_croc_soc/CTS_ccl_buf_00496/X sg13g2_buf_8 7 0.240 0.090 0.197 0.436 (509.28, 924.00)
i_croc_soc/CTS_ccl_a_buf_00486/X sg13g2_buf_16 32 0.321 0.060 0.101 0.537 (552.48, 922.32)
i_croc_soc/i_croc/CTS_ccl_a_buf_00331/X sg13g2_buf_16 32 0.147 0.039 0.098 0.634 (421.44, 904.68)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_168__reg/CLK sg13g2_dfrbp_1 - 0.147 0.043 0.008 0.642 (407.04, 873.60)
-----

```